

SNN Knowledge Distillation. We start from the original SNN Knowledge Distillation (KD) method. To illustrate the process of KD, we consider C-way classification task and denote $Z \in \mathbb{R}^C$ as the output of the network. Then, the class probability is given by:

$$Q_i = \frac{\exp(Z_i/\tau)}{\sum_{j=1}^C \exp(Z_j/\tau)}, \quad (4)$$

where Q_i and Z_i are the probability value on the i -th class and τ is the temperature scaling hyper-parameter. In knowledge distillation, τ is typically larger than 1.0, which helps control the flatness of the feature distribution to mitigate the phenomenon of overconfidence within the network. When τ equals 1.0, the output reverts to the vanilla Softmax output, denoted here as $S(\cdot)$ to represent this specific case. The objective of KD is to transfer knowledge from a high-performing teacher model to a lightweight student model. For the rescaled outputs, the original KD method achieves distillation by minimizing the KL divergence between the outputs of the teacher and student models.

$$KL(Q^{tea}||Q^{stu}) = \sum_{i=1}^C Q_i^{tea} \log\left(\frac{Q_i^{tea}}{Q_i^{stu}}\right), \quad (5)$$

where Q_i^{tea} and Q_i^{stu} indicates the probability value on the i -th category of the teacher and student output, respectively.

In addition, due to the potential errors in teachers' knowledge, KD also necessitates the use of cross-entropy to enable students to learn the true distribution of labels.

$$CE(Q^{stu}, y_{true}) = - \sum_{i=1}^C Q_i^{stu} \log(y_{true}) \quad (6)$$

where y_{true} denote the true labels.

The distinction between SNNs and ANNs lies in the fact that SNNs incorporate an additional temporal dimension T , which denotes the time step. Consequently, traditional SNN distillation attempts to adapt the distillation methods designed for ANNs by employing a simplistic approach that averages the outputs across the temporal dimension. By introducing a parameter α to control the weight ratio between true labels and soft labels. The loss calculation formula can be expressed as follows:

$$\begin{aligned} \mathcal{L}_{SKD} = & (1 - \alpha) \cdot CE\left(\frac{1}{T} \sum_{t=1}^T Q^{stu}(t), y_{true}\right) \\ & + \alpha \cdot KL(Q^{tea}||\frac{1}{T} \sum_{t=1}^T Q^{stu}(t)) \end{aligned} \quad (7)$$

3.2. Temporal Separation Knowledge Distillation with Entropy Regularization

In this section, we provide an in-depth explanation of the temporal separation knowledge distillation method with entropy regularization. Our approach consists of two core

components: temporal separation strategy and entropy regularization.

Temporal Separation Strategy. As illustrated in Fig. 1a, the original SNN distillation merely transferred the conventional KD methods to SNNs in a naive manner. The use of mean values to integrate outputs across the temporal dimension fails to adequately account for the unique spatiotemporal characteristics of SNNs. Due to the incremental temporal processing in SNNs, the output features at each time step exhibit variability and contain rich information. However, this also implies the potential presence of local outliers in the outputs at different time steps. While the averaging operation mitigates the impact of these anomalies, it poses risks for the subsequent model learning and probabilistic confidence. Consequently, implementing a temporal separation strategy for learning time step's output is crucial.

As shown in Fig. 1b, we apply temporal separation strategy to CE and KL in Eq. 7, moving the mean operation outside. We directly use the outputs at each time step to distill the learning of the true labels and the teacher distribution. The loss expression for temporal separation knowledge distillation is given by

$$\begin{aligned} \mathcal{L}_{TS} = & (1 - \alpha) \cdot \frac{1}{T} \sum_{t=1}^T CE(Q^{stu}(t), y_{true}) \\ & + \alpha \cdot \frac{1}{T} \sum_{t=1}^T KL(Q^{tea}||Q^{stu}(t)). \end{aligned} \quad (8)$$

Entropy Regularization. After introducing the temporal separation strategy, the soft labels from the teacher network directly impact the output distribution at each time step. This approach, however, can inadvertently reinforce erroneous information from the teacher network, increasing the likelihood that the student model will adopt incorrect knowledge. To counter this, we introduce an entropy regularization term, which constrains and adjusts the teacher's outputs to reduce excessive confidence in potentially erroneous knowledge. The formula for entropy is as follows:

$$H(Q^{stu}) = - \sum_{i=1}^C Q_i^{stu} \log(Q_i^{stu}). \quad (9)$$

Then, we obtain the loss formula for the entropy regularization term:

$$\mathcal{L}_{ER} = -\lambda \cdot \frac{1}{T} \sum_{t=1}^T H(Q^{stu}(t)), \quad (10)$$

where λ is the penalty factor.

Finally, we integrate the temporal separation strategy, as shown in Eq. 8, with the entropy regularization term from Eq. 10 to derive a new loss function as follows:

$$\mathcal{L}_{TSER} = \mathcal{L}_{TS} + \mathcal{L}_{ER}. \quad (11)$$

And Algorithm. 1 outlines the overall training process of our proposed method.

4. Experiment

In this section, we conduct extensive experiments to validate the effectiveness of our method. Initially, we compare our approach with existing SNN distillation methods. Furthermore, we conduct ablation studies, sensitivity analyses, energy consumption assessments, and visualizations to provide a comprehensive evaluation of our method.

4.1. Experimental Settings

Our proposed method is evaluated on three datasets. For teacher ANN networks, we select several advanced models, including ResNet-19 and ResNet-34 [21] as well as VGG-16. In contrast, for the student SNN models, we employ neural network architectures with equivalent or fewer layers, specifically ResNet-18/19, and VGG-11/16. Additionally, we conduct multiple experiments by varying the time step.

4.2. Comparison on Static Datasets

Our method is compared with previous approaches on CIFAR10 and CIFAR100, as shown in the Table. 1. The re-

Algorithm 1 Training student SNN model with TSER knowledge distillation for one epoch.

Require: pre-trained teacher ANN model M_t ; an initialized student SNN model M_s ; input dataset sample X ; and the true labels y_{true} ; total training iteration in one epoch I_{train} ; total validation iteration in one epoch I_{val}

Ensure: SNN model with TSER KD

for $i = 1$ to I_{train} **do**

 Get mini-batch training data, and true label: X^i, y_{true}^i ;
 Compute the M_s output $Z^{stu,i}(t)$ of each time step;
 Compute the M_t output $Z^{tea,i}$;
 Calculate loss function: $\mathcal{L}_{TSER}$;
 Backpropagation and update model parameters;

end for

for $i = 1$ to I_{val} **do**

 Get mini-batch validation data, and true label: X^i, y_{true}^i ;
 Compute the SNN average output $Z_{mean}^{stu,i} = \frac{1}{T} \sum_{t=1}^T Z^{stu,i}(t)$;
 Compare the $Z_{mean}^{stu,i}$ and y_{true}^i for classification;

end for

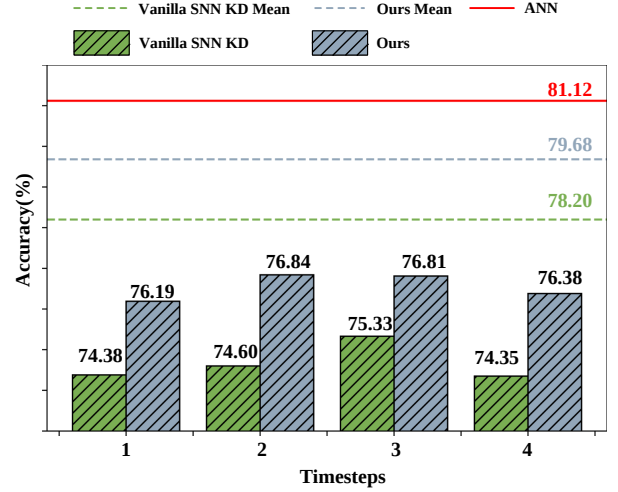


Figure 2. Prediction Accuracy Distribution at Different Time Steps for Vanilla SNN KD and Our Proposed Method. The solid red line indicates the teacher model’s accuracy, while the dashed lines represent the prediction accuracies of different distillation methods after averaging outputs over time steps. The bars show the prediction accuracies of each distillation method at individual time steps.

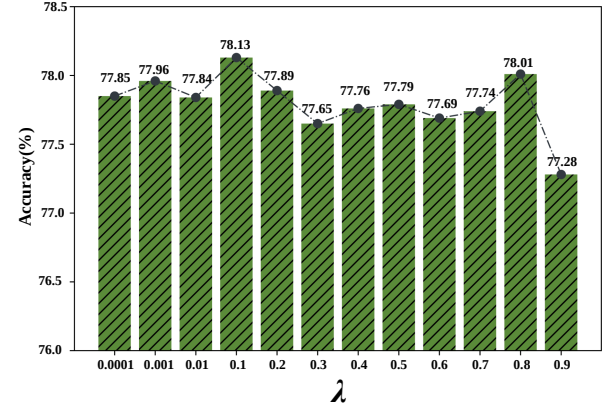


Figure 3. Accuracy Distribution for Different λ Values. Experiments conducted on the CIFAR100 dataset with a fixed time step of 2, testing various λ values.

sults demonstrate our method surpasses the SOTA performance achieved by prior method across both ResNet and VGG architectures. Specifically, ResNet19 outperform the previous SOTA by 0.57% and 0.41% on CIFAR10 and CIFAR100, respectively, achieving accuracies of 96.72% and 81.29%. This indicates that our approach, although based on logit distillation, performs exceptionally well compared to previous, more complex distillation methods. By addressing the limitations of simple transfer in existing distillation, our framework effectively learns from the teacher’s knowledge, minimizing the gap between the student and

	Method	Archi.	Acc.(%)		Archi.	Acc.(%)		Archi.	Acc.(%)		Archi.	Acc.(%)	
			CF-100	CF-10		CF-100	CF-10		CF-100	CF-10		CF-100	CF-10
without KD	Timestep=2												
	GLIF[55]	R-19	75.48	94.44	R-18	74.60	94.15	R-19	77.05	94.85	R-18	76.42	94.67
	TET[11]	R-19	72.87	94.16	-	-	-	R-19	74.47	94.44	-	-	-
	LSG[35]	R-19	76.32	94.41	-	-	-	R-19	76.85	95.17	-	-	-
	PFA[7]	R-19	76.70	95.60	-	-	-	R-19	78.10	95.71	-	-	-
	MPBN[18]	R-19	79.51	96.47	V-16	73.88	93.96	R-19	80.10	96.52	V-16	74.74	94.44
	IM-LIF[36]	R-19	77.21	95.29	-	-	-	R-19	77.42	95.66	-	-	-
	Spikformer[64]	-	-	-	-	-	-	S-4-256	75.96	93.94	S-4-384	77.86	95.19
	Spikingformer[61]	-	-	-	-	-	-	Sg-4-256	77.43	94.77	Sg-4-384	79.09	95.61
	SD Transformer [54]	-	-	-	-	-	-	S-2-512	78.40	95.60	-	-	-
with KD	Teacher ANN Timestep=1												
	Teacher ANN	R-34	81.12	97.10	V-16	78.08	96.06	R-34	81.12	97.10	V-16	78.08	96.06
		R-19	81.97	97.08	V-16	78.08	96.06	-	-	-	V-16	78.08	96.06
	Timestep=2												
	LaSNN [24]	R-18	76.17	93.64	V-16	73.80	93.90	R-18	78.12	95.09	V-16	74.99	94.49
		R-19	80.30	95.26	V-11	69.89	90.23	-	-	-	V-11	70.74	90.42
	KDSNN [50]	R-18	77.16	95.25	V-16	74.65	94.26	R-18	78.46	95.72	V-16	75.98	94.85
		R-19	80.88	96.15	V-11	72.17	92.81	-	-	-	V-11	73.05	92.91
	BKDSNN [51]	R-18	73.30	93.64	V-16	73.80	94.10	R-18	75.57	95.09	V-16	74.99	94.55
		R-19	75.74	95.26	V-11	69.89	92.73	-	-	-	V-11	70.74	92.88
	Ours	R-18	78.30 ± 0.08	95.58 ± 0.08	V-16	75.81 ± 0.12	94.55 ± 0.09	R-18	79.69 ± 0.02	96.18 ± 0.05	V-16	77.06 ± 0.04	95.01 ± 0.10
		R-19	81.29 ± 0.14	96.72 ± 0.02	V-11	73.67 ± 0.04	93.31 ± 0.02	-	-	-	V-11	74.65 ± 0.07	93.68 ± 0.02

Table 1. Comparison results with training-based SNN SOTA methods, including CNN- and transformer-based approaches, on CIFAR-10/100, with and without Knowledge Distillation (KD). **Acc.** denotes accuracy, **CF** denotes CIFAR, and **Archi.** denotes architecture. The abbreviations R, V, S, and Sg represent ResNet, VGG, Spikeformer, and Spikingformer architectures, respectively, while SD Transformer refers to the Spike-Driven Transformer. For KD methods, the student model aligns with the architecture of the teacher ANN.

teacher models. This reduction in disparity validates the superiority of our method.

Further comparisons on ImageNet dataset are detailed in Table. 2. The SEW ResNet-18 and SEW ResNet-34 architectures achieve top-1 accuracies of 69.24% and 73.16%, respectively. These results indicate that our method is capable of delivering competitive performance even within large-scale datasets, effectively harnessing the knowledge from ANNs to enhance the learning capabilities of SNNs.

4.3. Ablation Study

Temporal Separation Strategies and Entropy Regularization. We investigate different configurations of $\mathcal{L}_{TSER}$, focusing on the temporal separation applied within the KL and CE losses, as well as the integration of an entropy regularization term. The results are summarized in Table. 3. Our experiments reveal that incorporating tempo-

ral separation strategies notably enhances model accuracy, with improvements of 0.49% and 0.26% for the standard KL and CE loss function, respectively. When temporal separation is applied concurrently to both KL and CE losses, the overall accuracy increase reaches 0.84%. Furthermore, we find that optimally adjust the parameter λ in the entropy regularization term, alongside temporal separation, leads to an enhancement of up to 0.98%. These findings underscore the effectiveness of combining temporal separation and entropy regularization in maximizing the performance of SNN distillation.

Comparison of Output Accuracy at Different Time Steps. To illustrate the adaptability of our proposed distillation method for SNNs, we perform a comparative analysis of accuracy across different time steps between vanilla SNN KD and our method. As shown in Fig. 2, the accu-